Post-Quantum Secured, Hierarchical Group End-to-End Encryption for Multi-Consumer Industrial Sensor Networks

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Introduction

- Industrial IoT deployments use MQTT publish/subscribe to carry sensor telemetry from constrained field devices to multiple simultaneous consumers.
- A shared broker terminates every connection and routes every message under a hierarchical topic namespace (e.g. factory/line3/oven/temperature).
- Even when the client–broker link uses TLS, the broker is the endpoint of that session — it can structurally read or tamper with plaintext.
- Existing end-to-end (E2E) encryption for this setting is either strictly one-to-one, or a flat group key with no hierarchy across topics.

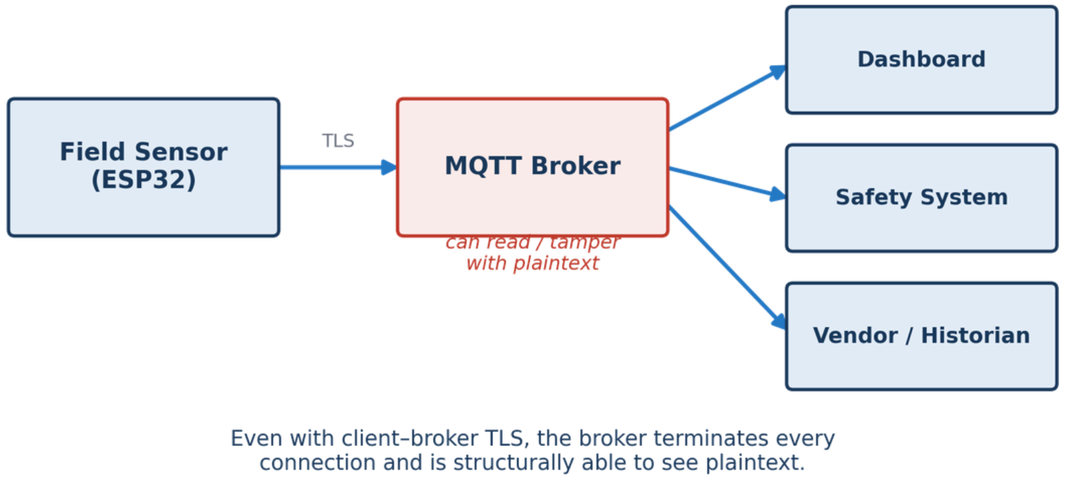


Fig. 1. Broker as an untrusted relay in MQTT-based IIoT telemetry

Introduction (contd.) — Motivation

- Application-layer encryption is necessary: broker infrastructure is often shared/third-party, so securing only the transport link is not sufficient.
- A flat, per-topic group key removes double-encryption overhead, but a subscriber needing an entire branch (e.g. all of line3) still needs one key per topic.
- Key-establishment handshakes in existing work rely on classical Diffie–Hellman, which is breakable by a sufficiently large quantum computer.
- Industrial control systems often stay in service for a decade or more — squarely inside the “harvest now, decrypt later” threat horizon.



Fig. 2. “Harvest now, decrypt later” threat to classical key-establishment

Literature Survey — Constrained-Hardware E2E Encryption

Amirkhanova et al., "A lightweight, end-to-end encrypted data pipeline for IIoT: An AES-GCM implementation for ESP32, MQTT, and Raspberry Pi", Information, Vol. 17, No. 1, 2026.

Key Points

- AES-256-GCM end-to-end encryption on ESP32 → broker → Raspberry Pi gateway, with the broker treated as untrusted.
- 41.754 ms average packet-preparation latency; 99.72% decrypt-and-store success over a 720-message soak test.
- Limitation: a single publisher, single pre-shared key, and single gateway only.

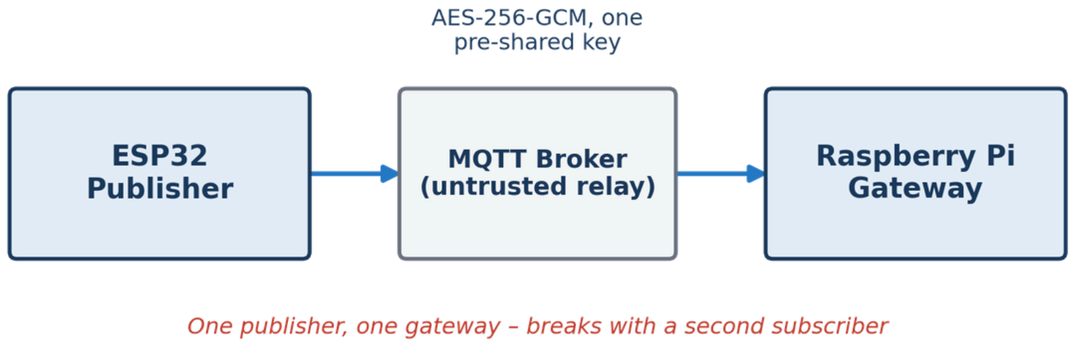


Fig. 3. One-to-one E2E encryption pipeline (Amirkhanova et al. [1])

Literature Survey (contd.) — Broker-Based Group-Wise E2E

Chien, Shih & Huang, "Exploring MQTT broker-based, end-to-end models for security and efficiency", Sensors, Vol. 25, No. 17, 2025.

Key Points

- Identifies double-encryption and broker decrypt-re-encrypt overhead in the conventional E2E-over-TLS layering.
- Best scheme: one shared group key per topic, broker relays ciphertext only; verified with AVISPA for up to 200 subscribers.
- Limitation: the group key is flat and scoped to exactly one topic — no branch-level access.

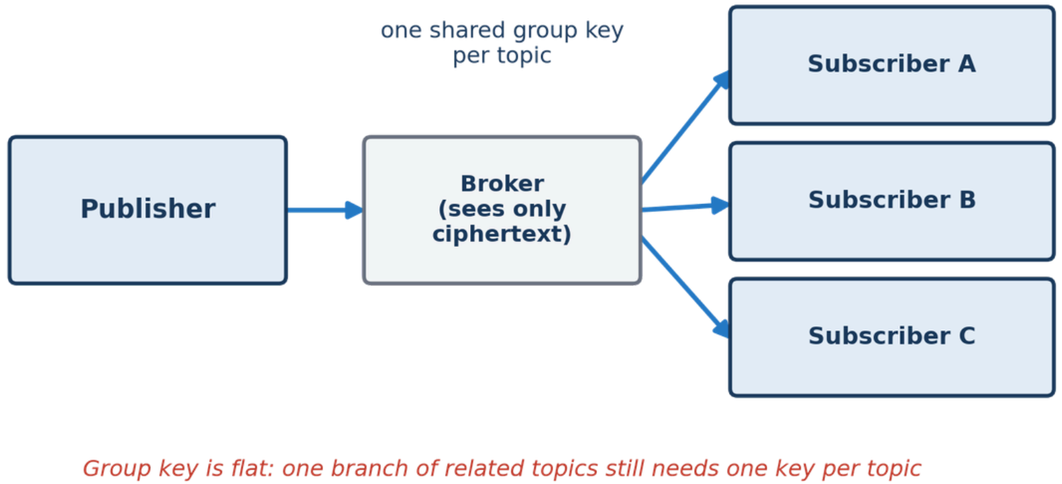


Fig. 4. Flat, per-topic group-key E2E model (Chien et al. [2])

Literature Survey (contd.) — Post-Quantum MQTT Handshakes

Malina, Dobias, Dzurenda & Srivastava, "Quantum-resistant and secure MQTT communication", ARES 2024; Samandari & Gritti, "Post-quantum authentication in the MQTT protocol", J. Cybersecurity & Privacy, 2023.

Key Points

- Replace classical Diffie–Hellman with an ML-KEM (Kyber) key encapsulation mechanism for the client–broker handshake.
- Shown to be practical on constrained and gateway-class hardware.
- Limitation: secures only a single client's handshake — no group or hierarchical setting considered.



Fig. 5. Post-quantum key-encapsulation handshake for MQTT [3,4]

Literature Survey (contd.) — Tree-Based Group Key Management

Wallner, Harder & Agee, "Key management for multicast: Issues and architectures", RFC 2627, 1999; Bertino et al., "Quantum-resistant networks: A review of primitives, protocols and best practices", arXiv:2605.04129, 2026.

Key Points

- Logical Key Hierarchy (LKH): members sit at the leaves of a key tree; a join/leave rekeys only the root-to-leaf path — O(log n), not O(n).
- Only the bootstrap handshake needs a PQC KEM; the tree itself stays purely symmetric .
- Limitation: this tree + PQC pattern has not been adapted to MQTT's topic hierarchy.

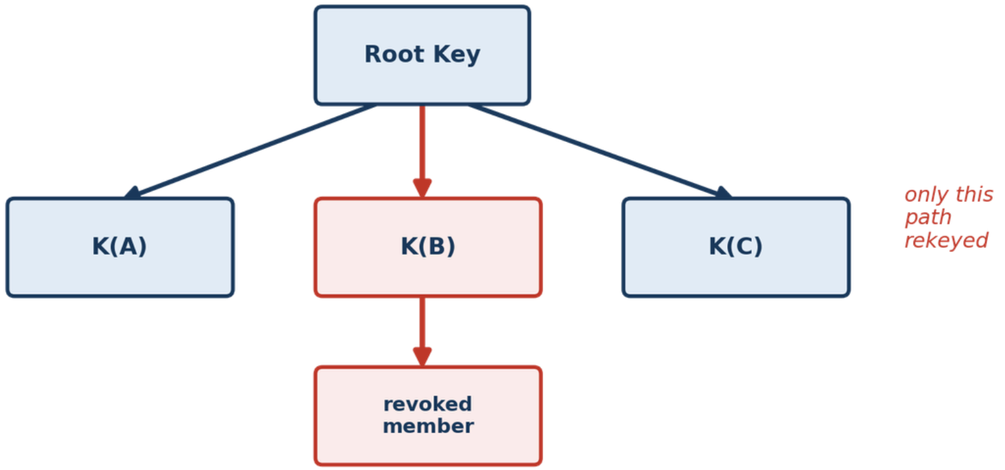
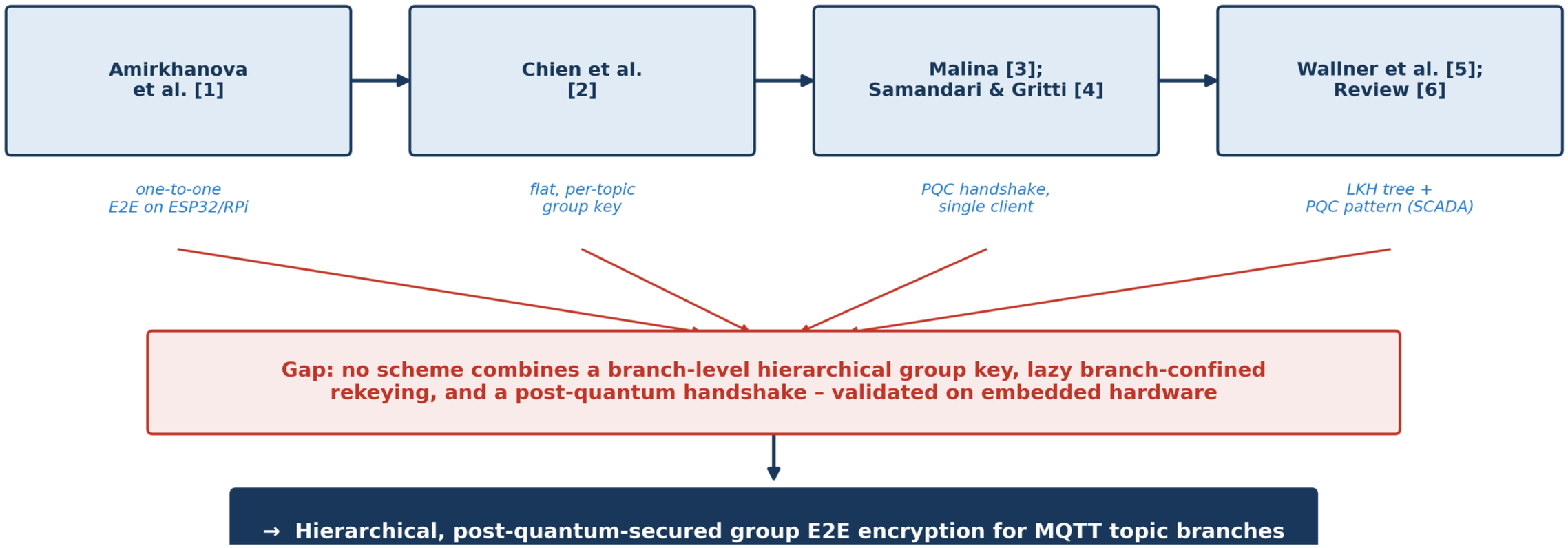


Fig. 6. Logical Key Hierarchy: lazy, path-confined rekeying [5,6]

Outcome of Literature Survey

- The six reviewed papers move along four fronts — payload encryption, group keying, PQC handshakes, and tree-based revocation — converging on one gap:



Problem Statement and Objectives of the Project

Problem Statement

- Existing E2E-encrypted MQTT schemes are either strictly one-to-one, or use a flat per-topic group key with no branch-level access — and both rely on a classical, quantum-vulnerable handshake.

Objectives

- Extend one-to-one E2E encryption to a hierarchical, group-wise scheme where one credential covers an entire topic branch.
- Derive each topic's key from its parent via a one-way KDF, so a single ancestor key yields every descendant key.
- Confine revocation and rekeying to the affected root-to-node path, triggered lazily on the next publish.
- Replace the classical Diffie–Hellman handshake with a post-quantum KEM (ML-KEM), keeping AES-256-GCM for payloads.

Methodology — High-Level System Architecture

- Extends the validated one-to-one ESP32 → broker → Raspberry Pi pipeline to multiple simultaneous publishers and subscribers under one shared topic branch.
- Every device first completes a post-quantum handshake with the broker to obtain its authorised branch's key material.
- Publishers then encrypt once with AES-256-GCM under a topic-derived key; the broker only relays ciphertext, never decrypting it.

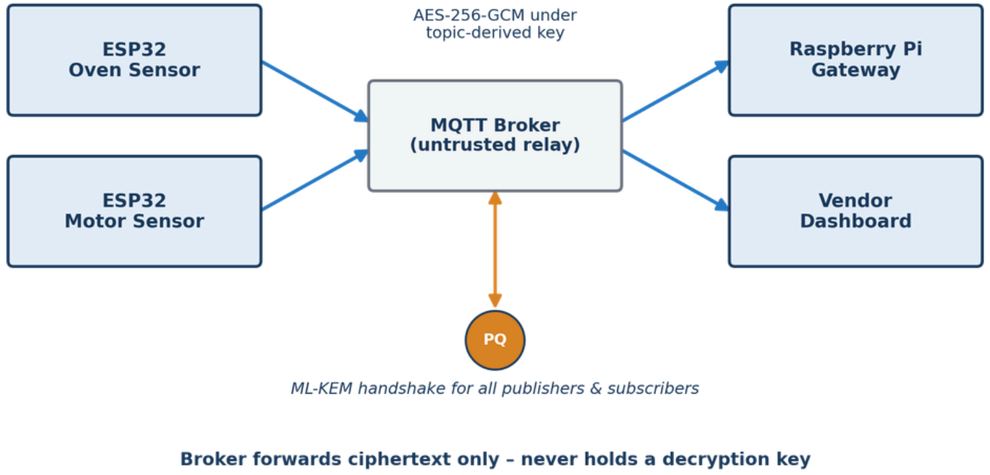


Fig. 7. Proposed hierarchical, PQ-secured group E2E architecture

Methodology — Topic-Hierarchical Key Derivation

- Each topic's key is derived from its parent's key with a one-way KDF: K(child) = KDF(K(parent), child_label).
- A device holding one ancestor key can compute every descendant key without any further exchange with the broker.
- Because the KDF is one-way, a leaf key cannot yield its parent's key or a sibling branch's key.

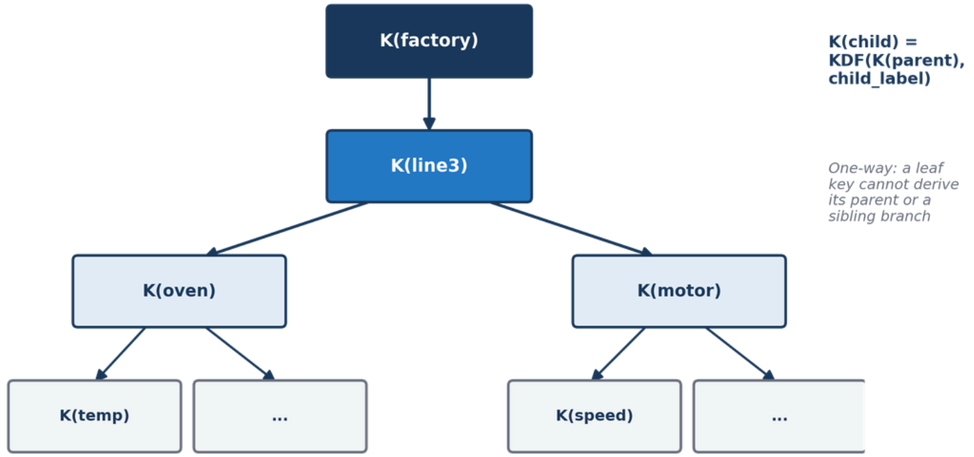


Fig. 8. Topic-hierarchical key derivation across a production line

Methodology — Lazy Rekeying & Post-Quantum Handshake

- Revocation follows Logical Key Hierarchy: only keys on the affected root-to-node path are regenerated, bounding cost to O(log n).
- Rekeying is lazily triggered — deferred until the next publish on that branch, with a bounded staleness timer as a backstop.
- The classical Diffie–Hellman handshake is replaced by ML-KEM only at bootstrap; per-message cost stays at the AES-GCM baseline.

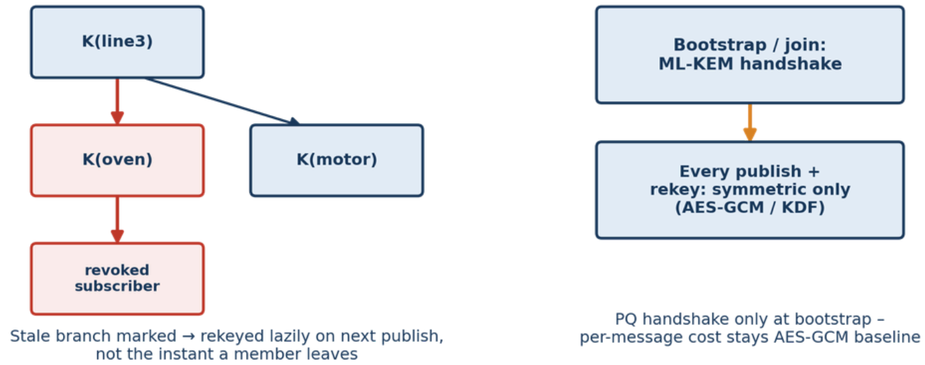


Fig. 9. Branch-confined lazy rekeying and post-quantum bootstrap

Conclusion and Future Work

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References

1. Gulshat Amirkhanova, Syrym Ismailov, Alikhan Amirkhanov, Saltanat Adilzhanova, Meiramkul Zhasuzakova, and Siming Chen, “A lightweight, end-to-end encrypted data pipeline for IIoT: An AES-GCM implementation for ESP32, MQTT, and Raspberry Pi”, Information, 17(1):33, 2026.
2. Hung-Yu Chien, An-Tong Shih, and Yuh-Ming Huang, “Exploring MQTT broker-based, end-to-end models for security and efficiency”, Sensors, 25(17):5308, 2025.
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1. D. Wallner, E. Harder, and R. Agee, “Key management for multicast: Issues and architectures”, RFC 2627, IETF, 1999.
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Timeline of the Project

Phase	Milestone	Status
Phase 1	Literature survey, problem formulation, and design of the proposed methodology	**Completed**
Phase 2	Implementation of the AES-GCM baseline pipeline on the ESP32 / Raspberry Pi testbed	**Planned**
Phase 3	Implementation of topic-hierarchical key derivation and tree-based lazy rekeying	**Planned**
Phase 4	Integration of the ML-KEM post-quantum handshake using liboqs	**Planned**
Phase 5	Benchmarking, comparison against baselines, and final report / defence	**Planned**

Literature Survey (contd.)

SLNo	Paper	Focus
1	Dynamic Access Control with End-to-End Message Security for MQTT	MQTT access control
2	Post-quantum cryptographic authentication protocol for industrial IoT using lattice-based cryptography	PQC authentication, industrial IoT